

Himanshu Yadav

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SUMMARY

Computer Science student interested in applied AI and ML systems. I build end-to-end NLP and LLM applications including vector retrieval pipelines, agent workflows, and structured inference systems. My work also focuses on the engineering side of machine learning - designing robust systems that integrate models with real infrastructure.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

AI/ML Systems: NLP, LLM Applications, Transformers (Attention Mechanisms), Embeddings, Vector Retrieval, Multi-Agent Systems, Prompt Engineering, Model Evaluation

Deep Learning: Neural Networks, RNNs, Supervised & Unsupervised Learning, Feature Engineering

Frameworks & Libraries: PyTorch, Hugging Face Transformers, Scikit-learn, FastAPI, Flask, Pydantic, SQLAlchemy, NumPy, Pandas

ML Systems: Inference Pipelines, System Debugging & Validation

Backend & Infrastructure: REST APIs, Async Processing, PostgreSQL, Vector Databases, Docker, Linux

Tools & Platforms: Git, GitHub, Render, Neon

PROJECTS

AegisAI - Self-Healing LLM Orchestration System | Project Link | *LLM Systems, RAG, AI Safety* April 2026

- Engineered LLM orchestration layer, reducing hallucination pass-through rate to 5% via structured validation.
- Built zero-trust security pipeline, achieving 100% block rate using hybrid LLM and heuristic detection.
- Designed multi-agent reasoning system, enabling 60% compositional query success via structured decomposition.
- Implemented self-healing RAG pipeline, improving reliability via dynamic query rewriting and recovery loops.

Transformer From Scratch - Mechanistic Attention Analysis | Project Link | *PyTorch* March 2026

- Built encoder-decoder Transformer from scratch, achieving 100% causal correctness with zero token leakage.
- Engineered stable attention computation, eliminating NaNs under fully masked conditions across all runs.
- Developed analysis pipeline to quantify head behavior using entropy, similarity, and positional metrics.
- Validated head importance via causal ablation, revealing identity heads as structural bottlenecks.

MeetSync - AI Meeting Minutes Generator | Live Demo | *Python, Flask, NLP* February 2026

- Built AI meeting minutes system converting audio into structured decisions, tasks, and deadlines.
- Processed 30-minute meetings in ~35s using Deepgram transcription and Gemini summarization pipeline.
- Ensured structured output reliability using Pydantic validation and PostgreSQL-backed schema enforcement.
- Reduced response latency to <1s using asynchronous background threads for LLM processing.

HACKATHONS & ACHIEVEMENTS

- Hack The Box Global Hackathon:** Selected as one of **4 teams** representing Galgotias University for the global cybersecurity hackathon hosted by YNOV Campus, Paris.
- Hackaccino 3.0 National Hackathon:** Developed a data-driven women's safety platform predicting area-wise risk levels from historical crime datasets; secured **17th place out of 194 teams**.
- NASA Space Apps Challenge:** Designed and built an interactive educational web platform in a team to simplify exoplanet science for ages 5-15, translating exoplanet concepts into engaging visual explanations.

EDUCATION

Galgotias University

B.Tech in Computer Science and Engineering

Gautam Buddha Nagar

2023 - 2027